Rekcod

Analyst: Benjamin Liu

Date: 4/12/2026

1.Executive Summary

This assessment analyzes several Docker images to identify filesystem artifacts, environment variables, layer metadata, and efficiency characteristics.The investigation involved:

- Enumerating available images
- Inspecting image layers and history
- Extracting filesystem paths and permissions
- Reviewing environment variables
- Evaluating image efficiency using *dive*

All findings are derived from direct command output contained in the uploaded document.

2.Investigation Screenshots

Docker reference： https://buddy.works/tutorials/docker-commands-cheat-sheet#list-all-images-including-dangling-images

jq reference : https://jqlang.org/tutorial/

How many docker images are there? *(7 points)*

11 Correct! ✔

```
ubuntu@ip-10-0-1-120:~$ sudo docker images -a
REPOSITORY                TAG          IMAGE ID       CREATED        SIZE
wordpress                 cli-php8.0   a0afbbe88d52   4 years ago    139MB
hhvm/hhvm                 latest       1ed16f91fd98   4 years ago    714MB
redis                     latest       08502081bff6   4 years ago    105MB
mysql                     latest       5c62e459e087   4 years ago    556MB
memcached                 latest       08c95b9d2533   4 years ago    82.4MB
ubuntu                    rolling      093da574621d   4 years ago    80.3MB
wagoodman/dive            latest       822b23d200a3   5 years ago    82.4MB
hello-world               latest       d1165f221234   5 years ago    13.3kB
dynod/call-make-action    1.0.0        2ad5efe096a8   5 years ago    264MB
dduportal/bats            1.1.0        95dc2c0854cd   6 years ago    11.1MB
explorer/fedena           2.3          112d84a07f3a   10 years ago   566MB
```

What directory is created in the Redis image, and what usergroup owns the directory? *(7 points)*

data, redis Correct! ✔

```
ubuntu@ip-10-0-1-120:~$ sudo docker history redis
IMAGE          CREATED       CREATED BY                                      SIZE      COMMENT
08502081bff6   4 years ago   /bin/sh -c #(nop)  CMD ["redis-server"]         0B
<missing>      4 years ago   /bin/sh -c #(nop)  EXPOSE 6379                  0B
<missing>      4 years ago   /bin/sh -c #(nop)  ENTRYPOINT ["docker-entry…   0B
<missing>      4 years ago   /bin/sh -c #(nop) COPY file:df205a0ef6e6df89…   374B
<missing>      4 years ago   /bin/sh -c #(nop) WORKDIR /data                 0B
<missing>      4 years ago   /bin/sh -c #(nop)  VOLUME [/data]               0B
<missing>      4 years ago   /bin/sh -c mkdir /data && chown redis:redis …   0B
<missing>      4 years ago   /bin/sh -c set -eux;   savedAptMark="$(apt-m…   31.7MB
<missing>      4 years ago   /bin/sh -c #(nop)  ENV REDIS_DOWNLOAD_SHA=ba…   0B
<missing>      4 years ago   /bin/sh -c #(nop)  ENV REDIS_DOWNLOAD_URL=ht…   0B
<missing>      4 years ago   /bin/sh -c #(nop)  ENV REDIS_VERSION=6.2.4      0B
<missing>      4 years ago   /bin/sh -c set -eux;  savedAptMark="$(apt-ma…   4.15MB
<missing>      4 years ago   /bin/sh -c #(nop)  ENV GOSU_VERSION=1.12        0B
<missing>      4 years ago   /bin/sh -c groupadd -r -g 999 redis && usera…   329kB
<missing>      4 years ago   /bin/sh -c #(nop)  CMD ["bash"]                 0B
<missing>      4 years ago   /bin/sh -c #(nop) ADD file:4903a19c327468b0e…   69.3MB
```

Where is the 'index.html' file in the image 'explorer/fedena'? What permissions does it have? *(7 points)*

./usr/share/doc/adduser/examples/adduser. Correct! ✔

`sudo docker run -it explore/fedena:2.3`

```
root@a1159277300a:/# find -iname "index.html"
./usr/lib/ruby/gems/1.8/doc/mysql-2.9.1/rdoc/index.html
./usr/lib/ruby/gems/1.8/doc/bundler-1.9.9/rdoc/index.html
./usr/lib/ruby/gems/1.8/doc/bundler-1.9.9/rdoc/classes/Bundler/Index.html
./usr/lib/ruby/gems/1.8/gems/rails-2.3.5/html/index.html
./usr/share/doc/adduser/examples/adduser.local.conf.examples/skel.other/index.ht
ml
./fedena-v2.3-bundle-linux/vendor/plugins/will_paginate/examples/index.html
```

```
root@a1159277300a:/# ls -l /usr/share/doc/adduser/examples/adduser.local.conf.ex
amples/skel.other/index.html
-rw-r--r-- 1 root root 1064 Oct 19  2011 /usr/share/doc/adduser/examples/adduser
.local.conf.examples/skel.other/index.html
```

chmod-calculator.com

Chmod Calculator

An awesome Chmod Calculator to convert Linux file permissions between different formats.

Owner	Group	Public
Read ☑	Read ☑	Read ☑
Write ☑	Write ☐	Write ☐
Execute ☐	Execute ☐	Execute ☐

Linux Permissions: 644 rw-r--r--

- Answer: ./usr/share/doc/adduser/examples/adduser.loca.conf.examples/skel.other/index.thml, 644

ctrl+q --> quit the container's interactive mode.

When is the creation date for the OS like layer in the 'dduportal/bats:1.1.0' image? *(7 points)*

1970-01-01T00:00:00z

Correct! ✔

Reference: https://medium.com/htc-research-engineering-blog/restore-rollback-layers-from-docker-image-a4e4b117d7e6

```
ubuntu@ip-10-0-0-195:~$ sudo mkdir newdir
ubuntu@ip-10-0-0-195:~$ sudo docker save -o newdir/myimage.tar dduportal/bats
ubuntu@ip-10-0-0-195:~$ ls
Desktop    Downloads  Pictures  Templates  newdir
Documents  Music      Public    Videos     snap
ubuntu@ip-10-0-0-195:~$ ls newdir/
myimage.tar
ubuntu@ip-10-0-0-195:~$ cd newdir/
ubuntu@ip-10-0-0-195:~/newdir$ tar -x ^C
ubuntu@ip-10-0-0-195:~/newdir$ tar -xf myimage.tar
tar: myimage.tar: Cannot open: Permission denied
tar: Error is not recoverable: exiting now
ubuntu@ip-10-0-0-195:~/newdir$ sudo tar -xf myimage.tar
ubuntu@ip-10-0-0-195:~/newdir$ ls
099c7bdd4ef25a3324403469459fa9ce7e2a86c95835f43984a3ebc54459893e
19300014ebb75fd617ed4843c906f1b0c6ff3c83d923637281a220609d6a941c
51b846330907e0af0496e649e3a78d46efab4f7c54a5aef24a642b925d5bbf72
8097754055649cdc13c2db5b031f7a0c9bda9f899bd0023ec5d0967d4648c34a
82fb445a041c10fc5c2c2e1cd5919bdc827d803dc4ca9dabe9bd18fb0bfa8e85
95dc2c0854cdf4d8bc19aaa1a159422d71a83be6d2fa881c491eac05cde57ecb.json
a61298a1d179786c2c176dc6c3e20c2d0cad25582e53a7bbec428fc4370a679e
fe2d6ad5109cf6ccfacebd4accd1b02f9182ea6dc02f694112016850493a7713
manifest.json
myimage.tar
repositories
```

```
ubuntu@ip-10-0-0-195:~/newdir$ jq '.' mainfest.json
jq: error: Could not open file mainfest.json: No such file or directory
ubuntu@ip-10-0-0-195:~/newdir$ jq '.' manifest.json
[
  {
    "Config": "95dc2c0854cdf4d8bc19aaa1a159422d71a83be6d2fa881c491eac05cde57ecb.json",
    "RepoTags": [
      "dduportal/bats:1.1.0"
    ],
    "Layers": [
      "a61298a1d179786c2c176dc6c3e20c2d0cad25582e53a7bbec428fc4370a679e/layer.tar",
      "51b846330907e0af0496e649e3a78d46efab4f7c54a5aef24a642b925d5bbf72/layer.tar",
      "8097754055649cdc13c2db5b031f7a0c9bda9f899bd0023ec5d0967d4648c34a/layer.tar",
      "099c7bdd4ef25a3324403469459fa9ce7e2a86c95835f43984a3ebc54459893e/layer.tar",
      "82fb445a041c10fc5c2c2e1cd5919bdc827d803dc4ca9dabe9bd18fb0bfa8e85/layer.tar",
      "fe2d6ad5109cf6ccfacebd4accd1b02f9182ea6dc02f694112016850493a7713/layer.tar",
      "19300014ebb75fd617ed4843c906f1b0c6ff3c83d923637281a220609d6a941c/layer.tar"
    ]
  }
]
```

```
ubuntu@ip-10-0-0-195:~/newdir/a61298a1d179786c2c176dc6c3e20c2d0cad25582e53a7bbec428fc4370a679e$ jq '.' json
{
  "id": "a61298a1d179786c2c176dc6c3e20c2d0cad25582e53a7bbec428fc4370a679e",
  "created": "1970-01-01T00:00:00Z",
  "container_config": {
    "Hostname": "",
    "Domainname": "",
    "User": "",
    "AttachStdin": false,
    "AttachStdout": false,
    "AttachStderr": false,
    "Tty": false,
    "OpenStdin": false,
    "StdinOnce": false,
    "Env": null,
    "Cmd": null,
    "Image": "",
    "Volumes": null,
    "WorkingDir": "",
    "Entrypoint": null,
    "OnBuild": null,
    "Labels": null
  },
  "os": "linux"
}
ubuntu@ip-10-0-0-195:~/newdir/a61298a1d179786c2c176dc6c3e20c2d0cad25582e53a7bbec428fc4370a679e$
```

The `docker image inspect` command provides detailed, low-level information about one or more Docker images in a JSON format. It is essential for debugging image configurations, verifying security settings (like environment variables or labels), and understanding the underlying layers.

What is the value set for the variable 'WORDPRESS_CLI_GPG_KEY' in the 'wordpress:cli-php8.0' image? *(7 points)*

63AF7AA15067C05616FDDD88A3A2E8F226 Correct! ✔

```
ubuntu@ip-10-0-15-231:~$ sudo docker image inspect wordpress:cli-php8.0
[
    {
        "Id": "sha256:a0afbbe88d52535772f3811e22b8994e17ce04b1f5797bbb6fb1e0b0473c320e",
        "RepoTags": [
            "wordpress:cli-php8.0"
        ],
        "RepoDigests": [
            "wordpress@sha256:51a6d9dc8f42bf6aab8172796e239c72bb7db9397d8538b2faa23068d5a44e2c"
        ],
        "Parent": "",
        "Comment": "",
        "Created": "2021-07-02T02:19:39.300848228Z",
        "Container": "bda5bd628836ab81ec41c7e1f949d47aff3af1c5803ef77f0b3ca89d7abfde9a",
        "ContainerConfig": {
            "Hostname": "bda5bd628836",
            "Domainname": "",
            "User": "www-data",
            "AttachStdin": false,
            "AttachStdout": false,
            "AttachStderr": false,
            "Tty": false,
            "OpenStdin": false,
            "StdinOnce": false,
            "Env": [
                "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
                "PHPIZE_DEPS=autoconf \t\tdpkg-dev dpkg \t\tfile \t\tg++ \t\tgcc \t\tlibc-dev \t\tmake \t\tpkgconf \t\tre2c",
```

```
            ],
            "OnBuild": null,
            "Labels": {}
        },
        "DockerVersion": "20.10.7",
        "Author": "",
        "Config": {
            "Hostname": "",
            "Domainname": "",
            "User": "www-data",
            "AttachStdin": false,
            "AttachStdout": false,
            "AttachStderr": false,
            "Tty": false,
            "OpenStdin": false,
            "StdinOnce": false,
            "Env": [
                "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
                "PHPIZE_DEPS=autoconf \t\tdpkg-dev dpkg \t\tfile \t\tg++ \t\tgcc \t\tlibc-dev \t\tmake \t\tpkgconf \t\tre2c",
                "PHP_INI_DIR=/usr/local/etc/php",
                "PHP_CFLAGS=-fstack-protector-strong -fpic -fpie -O2 -D_LARGEFILE_SOURCE -D_FILE_OFFSET_BITS=64",
                "PHP_CPPFLAGS=-fstack-protector-strong -fpic -fpie -O2 -D_LARGEFILE_SOURCE -D_FILE_OFFSET_BITS=64",
                "PHP_LDFLAGS=-Wl,-O1 -pie",
                "GPG_KEYS=1729F83938DA44E27BA0F4D3DBDB397470D12172 BFDDD28642824F8118EF77909B67A5C12229118F",
                "PHP_VERSION=8.0.8",
                "PHP_URL=https://www.php.net/distributions/php-8.0.8.tar.xz",
                "PHP_ASC_URL=https://www.php.net/distributions/php-8.0.8.tar.xz.asc",
                "PHP_SHA256=dc1668d324232dec1d05175ec752dade92d29bb3004275118bc3f7fc7cbfbb1c",
                "WORDPRESS_CLI_GPG_KEY=63AF7AA15067C05616FDDD88A3A2E8F226F0BC06",
```

What is the 'efficiency score' and 'wasted space' for the 'hhvm/hhvm' image according to the (no so secret) tool? *(8 points)*

99, 9

Correct! ✔

reference：https://github.com/wagoodman/dive

```
ubuntu@ip-10-0-15-231:~$ alias dive="sudo docker run -ti --rm -v /var/run/docker.sock:/var/run/docker.sock docker.io/wagoodman/dive"
ubuntu@ip-10-0-15-231:~$ dive hhvm/hhvm
Image Source: docker://hhvm/hhvm
Fetching image... (this can take a while for large images)
Analyzing image...
Building cache...
```

```
│ ● Layers ├──────────────────────────────────────────────────
Cmp   Size  Command
      73 MB  FROM 679f8666b5733b3
     641 MB  |2 HHVM_PACKAGE=hhvm=4.116.0-* HHVM_REPO_DISTRO=focal-4.

│ Layer Details ├──────────────────────────────────────────────

Tags:   (unavailable)
Id:     679f8666b5733b30fd18c673e22888dfa483b588add0a0b0b36d4c46d716
Digest: sha256:feef05f055c989eea0367f5d3a2cba79896dd6d8a8b72bea2c729
Command:
#(nop) ADD file:920cf788d1ba88f76c97e41e03e4dc2f3005b08d65b5e9da9dd1

│ Image Details ├──────────────────────────────────────────────

Image name: hhvm/hhvm
Total Image size: 714 MB
Potential wasted space: 9.0 MB
Image efficiency score: 99 %

Count   Total Space  Path
    2        3.9 MB  /usr/lib/x86_64-linux-gnu/libstdc++.so.6.0.28
    2        1.4 MB  /usr/lib/x86_64-linux-gnu/libsystemd.so.0.28.0
    2        1.3 MB  /var/cache/debconf/templates.dat
    2        812 kB  /var/lib/dpkg/info/libstdc++6:amd64.symbols
    2        447 kB  /var/log/dpkg.log
    2        415 kB  /var/lib/dpkg/status
    2        210 kB  /usr/lib/x86_64-linux-gnu/libgcc_s.so.1
    2        141 kB  /usr/share/doc/gcc-10-base/copyright
    2         60 kB  /var/log/lastlog
    2         50 kB  /var/lib/dpkg/info/libsystemd0:amd64.symbols
    2         47 kB  /var/log/apt/history.log
    2         24 kB  /etc/ld.so.cache
    2         22 kB  /usr/share/doc/libsystemd0/copyright
    2         20 kB  /var/cache/debconf/config.dat
    2         20 kB  /var/cache/ldconfig/aux-cache
    2         20 kB  /var/log/apt/eipp.log.xz
    2         14 kB  /var/lib/apt/extended_states
    2         12 kB  /var/log/alternatives.log
    2         11 kB  /usr/share/doc/libsystemd0/changelog.Debian.gz
```

```
│ Current Layer Contents ├──────────────────────────────────────
Permission     UID:GID       Size  Filetree
-rwxrwxrwx         0:0        0 B  ├── bin → usr/bin
drwxr-xr-x         0:0        0 B  ├── boot
drwxr-xr-x         0:0        0 B  ├── dev
drwxr-xr-x         0:0      122 kB ├── etc
-rw-------         0:0        0 B  │   ├── .pwd.lock
-rw-r--r--         0:0      3.0 kB │   ├── adduser.conf
drwxr-xr-x         0:0      100 B  │   ├── alternatives
-rw-r--r--         0:0      100 B  │   │   ├── README
-rwxrwxrwx         0:0        0 B  │   │   ├── awk → /usr/bin/mawk
-rwxrwxrwx         0:0        0 B  │   │   ├── nawk → /usr/bin/mawk
-rwxrwxrwx         0:0        0 B  │   │   ├── pager → /bin/more
-rwxrwxrwx         0:0        0 B  │   │   ├── rmt → /usr/sbin/rmt-t
-rwxrwxrwx         0:0        0 B  │   │   └── w → /usr/bin/w.procps
drwxr-xr-x         0:0       12 kB │   ├── apt
drwxr-xr-x         0:0      2.0 kB │   │   ├── apt.conf.d
-rw-r--r--         0:0       92 B  │   │   │   ├── 01-vendor-ubuntu
-rw-r--r--         0:0      630 B  │   │   │   ├── 01autoremove
-r--r--r--         0:0      623 B  │   │   │   ├── 01autoremove-kern
-rw-r--r--         0:0      182 B  │   │   │   ├── 70debconf
-rw-r--r--         0:0       44 B  │   │   │   ├── docker-autoremove
-rw-r--r--         0:0      318 B  │   │   │   ├── docker-clean
-rw-r--r--         0:0       70 B  │   │   │   ├── docker-gzip-index
-rw-r--r--         0:0       27 B  │   │   │   └── docker-no-languag
drwxr-xr-x         0:0        0 B  │   │   ├── auth.conf.d
drwxr-xr-x         0:0        0 B  │   │   ├── preferences.d
-rw-r--r--         0:0      2.7 kB │   │   ├── sources.list
drwxr-xr-x         0:0        0 B  │   │   ├── sources.list.d
drwxr-xr-x         0:0      7.3 kB │   │   └── trusted.gpg.d
-rw-r--r--         0:0      2.8 kB │   │       ├── ubuntu-keyring-20
-rw-r--r--         0:0      2.8 kB │   │       ├── ubuntu-keyring-20
-rw-r--r--         0:0      1.7 kB │   │       └── ubuntu-keyring-20
-rw-r--r--         0:0      2.3 kB │   ├── bash.bashrc
-rw-r--r--         0:0      367 B  │   ├── bindresvport.blacklist
drwxr-xr-x         0:0      201 B  │   ├── cron.d
-rw-r--r--         0:0      201 B  │   │   └── e2scrub_all
drwxr-xr-x         0:0      2.7 kB │   ├── cron.daily
-rwxr-xr-x         0:0      1.5 kB │   │   ├── apt-compat
-rwxr-xr-x         0:0      1.2 kB │   │   └── dpkg
```

3.Conclusion

The Docker images examined exhibit expected behaviors for legacy application stacks.Key observations include:

- Proper directory creation and ownership in Redis
- Legacy documentation files in explorer/fedena
- A Unix epoch timestamp in dduportal/bats (common for minimal base layers)
- Clear environment variable configuration in WordPress CLI
- High efficiency in hhvm/hhvm with minimal wasted space

These findings demonstrate correct use of Docker inspection, layer analysis, and filesystem enumeration techniques.